INSTITUTE OF TECHNOLOGY OF CAMBODIA

DEPARTMENT : APPLIED MATHEMATIC AND STATISTICS

SUBJECT : PROBABLISTIC GRAPHICAL MODELS

TOPIC : POS Tagger

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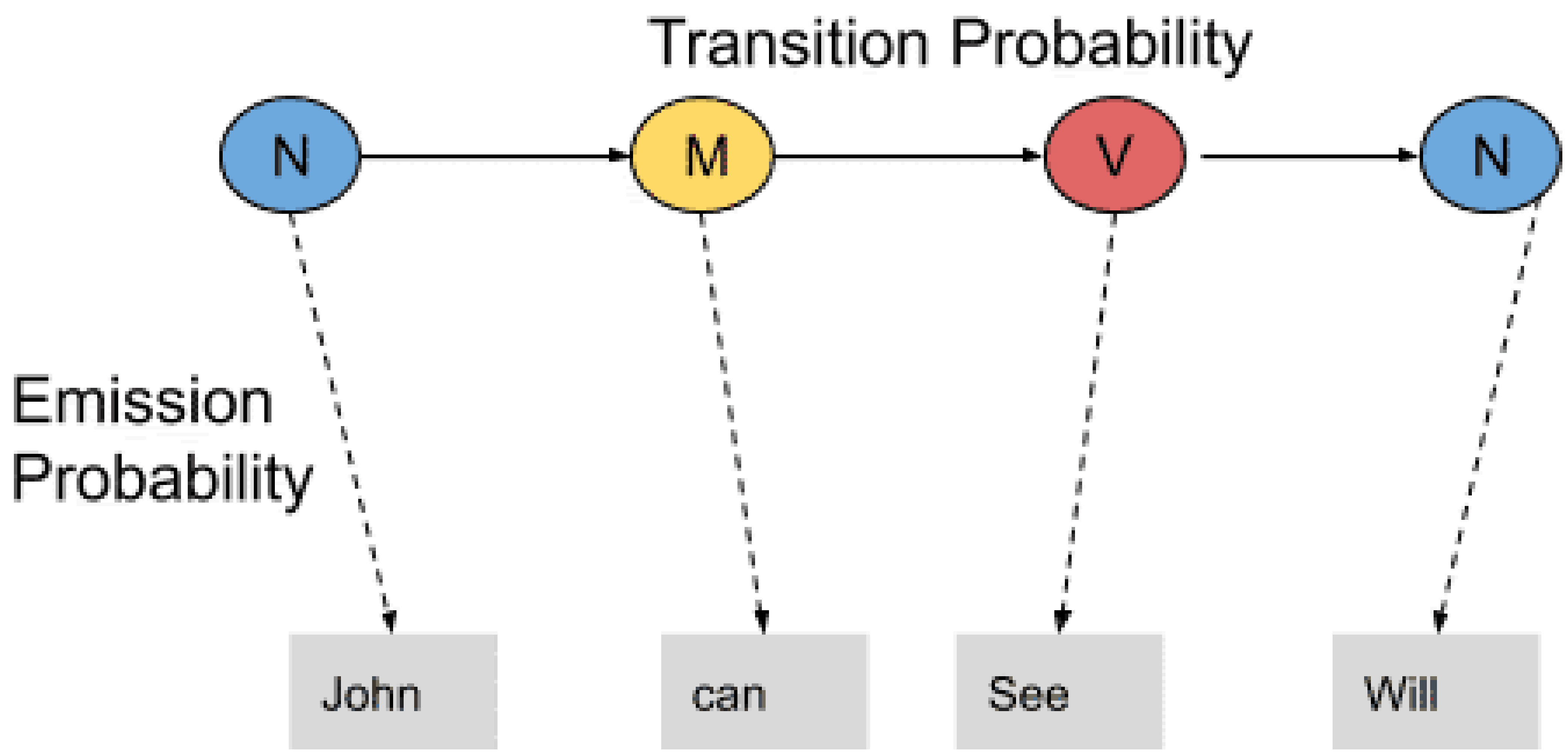
CONTENT

THE PROBLEM

OR

- POS tagging is difficult because many words are ambiguous.
- For example, 'watch' can be a noun or a verb depending on the context.
- Our goal was to build a system that can accurately label these words using probabilistic state transitions.
- Dataset: Brown Corpus

THE HMM APPROACH



Estimating three core parameters via Maximum Likelihood Estimation:

- Initial probabilities: the likelihood of a sentence starting with a specific POS tag
- Transition probabilities between tags (tag to tag)
- Emission probabilities of words from those tags (tag to word)

ENGINEERING HURDLES

Laplace Smoothing (OOV)

- Add-α-Smoothing: α=1.0
- To handle the Out-of-Vocabulary

$$P_{\text{smoothed}}(o_t|s_i) = \frac{\text{Count}(o_t \text{ tagged as } s_i) + \alpha}{\text{Count}(s_i) + \alpha|V|}$$

Log-Space Viterbi (Underflow)

- convert the Viterbi decoding algorithm into log-space
- add epsilon = 1e-12 to prevent log(0)

$$\log v_t(j) = \max_{i=1}^{N} \left[\log v_{t-1}(i) + \log P(s_j|s_i) \right] + \log P(o_t|s_j)$$

RESULT & EVALUATION

92.47%Word-Level Accuracy on the 20% test set

ERROR CASE STUDY

Sentence: What’s below the water-line inerests me also.

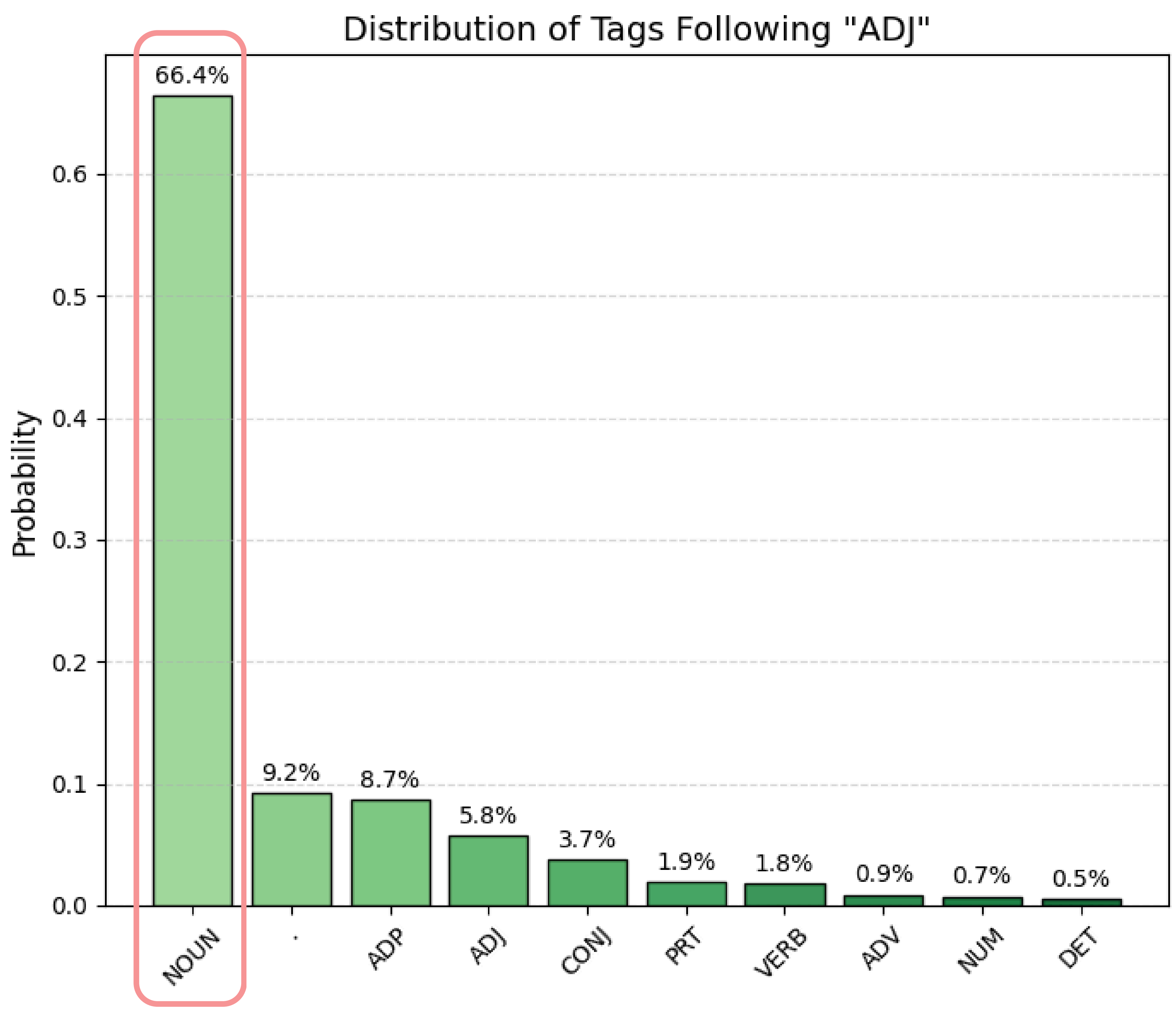
Word	True	Pred
What’s	PRT	PRT
below	ADP	ADP
the	DET	DET
water-line	NOUN	ADJ
interests	VERB	NOUN
me	PRON	PRON
also	ADV	ADV
.	.	.

LIMITATION

Sentence: What's below the water-line inerests me also.

$$\text{water-line} \xrightarrow{\text{First-Order Markov Assumption}} \text{Interest}$$

- Sentence: What's | below | the | water-line | interests | me | also .
- True: PRT | ADP | DET | NOUN | VERB | PRON | ADV | .
- Pred: PRT | ADP | DET | ADJ | NOUN | PRON | ADV | .



SUMMARY

n conclusion, while HMMs provide a highly effective, fast baseline for sequential tagging, their strict First-Order limitation makes them vulnerable to cascading errors on complex syntax.

Thank you